

FileEditSelectionViewGoRunTerminalHelp

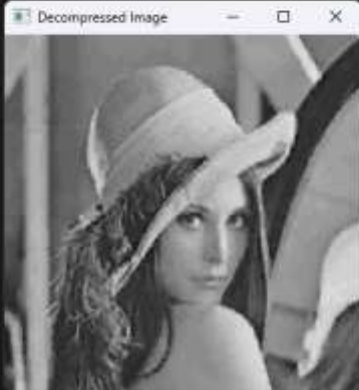
11402_1_D1349111.py M X11402_2_D1349111.py M11402_3_D1349111.py M11402_4_D1349111.py M

11402_1_D1349111.py > -
You 28 minutes ago | 1 author (You)
1 import cv2
2 import numpy as np
3 import pandas as pd
4 import math
5
6 data = pd.read_excel('codebook1.xlsx', header=None)
7 codebook = data.values
8
9 img = cv2.imread('lena.jpg', cv2.IMREAD_GRAYSCALE)
10 h, w = img.shape
11
12 index = np.zeros((h // 4, w // 4), dtype=np.uint8)
13 decompressed_img = np.zeros((h, w), dtype=np.uint8)
14
15 # 壓縮
16 for i in range(0, h, 4):
17 for j in range(0, w, 4):
18 block = img[i:i+4, j:j+4].flatten()
19
20 d = np.linalg.norm(codebook - block, axis=1)
21
22 best_index = np.argmin(d)
23
24 index[i//4, j//4] = best_index
25
26 # 解壓縮
27 for i in range(h // 4):
28 for j in range(w // 4):
29 idx = index[i, j]
30
31 codeword = codebook[idx]
32
33 decompressed_img[i*4:(i+1)*4, j*4:(j+1)*4] = codeword.reshape((4, 4))
34
35 psnr = cv2.PSNR(img, decompressed_img)
36
37 print(f'原圖與解壓縮比較的影像品質 PSNR: {psnr:.2f}')
38
39 index_img = cv2.normalize(index, None, 0, 255, cv2.NORM_MINMAX, dtype=cv2.CV_8U)
40
41 cv2.imshow('Original Image', img)
42 cv2.imshow('Compressed Index Image', index_img)
43 cv2.imshow('Decompressed Image', decompressed_img)
44
45 cv2.waitKey(0)
46 cv2.destroyAllWindows()
47

Original Image



Decompressed Image



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原圖與解壓縮比較的影像品質 PSNR: 29.14
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